

I3DION Spatial

Technical Architecture & System Design Document (TAD)

Version 1.0

1. Architecture Overview

Introduction

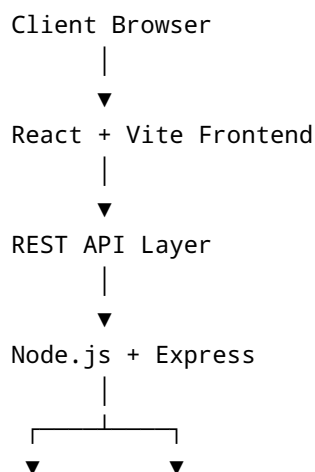
I3DION Spatial is designed as a modern web-based Industrial AR Catalog and Sales Visualization Platform.

The system architecture follows a modular, scalable, and cloud-ready approach that allows future expansion without requiring major architectural changes.

The architecture is divided into six primary layers:

1. Presentation Layer
 2. Application Layer
 3. API Layer
 4. Data Layer
 5. Storage Layer
 6. Deployment Layer
-

2. High-Level Architecture



```
Database    Storage
(Postgres) (MinIO)
  |
  ▼
AR & 3D Services
(Three.js + WebXR)
```

3. Frontend Architecture

Technology Stack

React

React will be used as the primary frontend framework.

Reason:

- Component Based Architecture
 - Large Ecosystem
 - Maintainability
 - Scalability
-

Vite

Used as the build tool.

Benefits:

- Faster Development
 - Lightweight Build Process
 - Modern Development Experience
-

Tailwind CSS

Used for UI styling.

Benefits:

- Faster UI Development
- Responsive Design
- Consistent Design System

Frontend Modules

Authentication Module

Responsibilities:

- Login
- Signup
- Session Handling

Dashboard Module

Responsibilities:

- Product Overview
- Catalog Overview
- Analytics

Product Management Module

Responsibilities:

- Create Products
- Edit Products
- Asset Upload

Catalog Builder Module

Responsibilities:

- Catalog Creation
- Catalog Editing
- QR Publishing

Product Experience Module

Responsibilities:

- Product Page
- Product Viewer
- AR Launch

4. 3D Visualization Layer

Technology

Three.js

Three.js will be the primary 3D rendering engine.

Responsibilities

Model Rendering

Render GLB models inside browser.

Camera Controls

Provide:

- Orbit
 - Zoom
 - Pan
-

Lighting System

Support:

- Ambient Light
 - Directional Light
 - HDR Environment Maps
-

Animation Support

Support:

- GLTF Animations
 - Product Demonstrations
 - Exploded Views
-

Supported Formats

V1

- GLB

Future

- GLTF
 - FBX
 - USDZ
-

5. AR Architecture

Technology

WebXR

Primary AR Framework.

Model Viewer

Used for simplified AR implementation.

Responsibilities

Device Detection

Check AR Support.

Surface Detection

Detect floors and placement surfaces.

Object Placement

Allow product placement.

Real Scale Visualization

Display products at actual dimensions.

Supported Platforms

Android

- Chrome
 - ARCore
-

iOS

- Safari
 - Quick Look
-

6. Product Lock System Architecture

Purpose

Maintain object position in world space.

Workflow

```
Place Product
↓
Create World Anchor
↓
Lock Position
↓
Track User Movement
↓
Keep Object Fixed
```

Components

Anchor Manager

Maintains world coordinates.

Position Controller

Controls:

- Lock
 - Unlock
 - Reset
-

Tracking Manager

Tracks user position changes.

7. Backend Architecture

Technology

Node.js

Runtime Environment.

Express.js

Backend Framework.

Responsibilities

Authentication

User Management.

Product Services

Product CRUD Operations.

Catalog Services

Catalog CRUD Operations.

QR Services

QR Generation.

Lead Services

Lead Management.

Analytics Services

Event Tracking.

8. API Architecture

API Style

REST API

Base Structure

```
/api/v1
```

Product APIs

```
GET /products
```

```
POST /products
```

PUT /products/:id

DELETE /products/:id

Catalog APIs

GET /catalogs

POST /catalogs

PUT /catalogs/:id

QR APIs

POST /qr/generate

Lead APIs

POST /lead/create

GET /lead/list

9. Database Architecture

Technology

PostgreSQL

Primary Database.

Why PostgreSQL

- Open Source
- Reliable

- ACID Compliance
 - Scalable
-

Core Tables

Users

Stores platform users.

Companies

Stores company information.

Products

Stores product information.

Catalogs

Stores catalog information.

Product Assets

Stores images, videos, and models.

QR Codes

Stores generated QR references.

Leads

Stores inquiries.

Analytics

Stores tracking events.

10. Storage Architecture

Technology

MinIO

Open-source object storage.

Stored Assets

Product Images

PNG JPG

Product Videos

MP4

Product Models

GLB

Documents

PDF

Folder Structure

```
storage/  
  companies/  
  products/  
  images/  
  videos/  
  models/
```

11. Security Architecture

Authentication

JWT Authentication.

Authorization

Role-Based Access Control.

Roles

Admin

Manager

Sales User

Customer

Data Protection

- Password Hashing
 - Secure Tokens
 - API Validation
-

12. Deployment Architecture

Development Environment

Local Development

Frontend
localhost:5173

Backend
localhost:5000

Production Environment

VPS Deployment

Ubuntu Server

Docker Containers

Frontend Container

Backend Container

Database Container

MinIO Container

Reverse Proxy

Nginx

SSL

Let's Encrypt

13. Monitoring & Logging

Application Logs

Backend Logs

Error Tracking

API Errors

Analytics

Track:

- Product Views
 - Catalog Views
 - QR Scans
 - AR Sessions
-

14. Scalability Strategy

Phase 1

Single Server Deployment

Phase 2

Separate Frontend & Backend

Phase 3

Dedicated Database Server

Phase 4

Multi-Tenant SaaS Architecture

15. Future Architecture Extensions

AI Recommendation Engine

Product Suggestions

Space Validation Engine

Fitment Analysis

Product Configurator

Color & Material Changes

Mobile Application

React Native

Offline PWA

Local Model Caching

Architecture Principles

The architecture of I3DION Spatial is designed around four core principles:

Simplicity

Easy to develop and maintain.

Scalability

Able to grow with future business needs.

Performance

Optimized for mobile-first product experiences.

Extensibility

Supports future AR, AI, analytics, and enterprise modules without major redesign.

Final Technical Definition

I3DION Spatial is a web-first Industrial AR Catalog & Sales Visualization Platform built using React, Vite, Three.js, WebXR, Node.js, PostgreSQL, and MinIO, designed to provide scalable, immersive, and interactive product experiences for industrial sales and marketing teams.